

# Portfolio Optimization with Black-Litterman

## A Case Study in Public Portfolio Disclosures

Walk-forward backtest · Real SEC & congressional filing data · Buffett · Pelosi · Trump

**Abstract.** This report describes a Python codebase that applies a Black-Litterman portfolio overlay to publicly disclosed equity holdings and evaluates it in a rolling walk-forward backtest. Three case studies are constructed from real regulatory filings — Berkshire Hathaway SEC 13F (Warren Buffett), U.S. House Financial Disclosure (Nancy Pelosi), and OGE 278e (Donald Trump) — and three strategies are compared: a static disclosed-weight portfolio, a sample mean-variance optimiser, and a Black-Litterman posterior fed into a long-only quadratic programme. The framework is fully YAML-configurable, ships with 63 unit and integration tests, and produces four analysis notebooks covering data quality, strategy comparison, hyperparameter sensitivity, and benchmark attribution.

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# 1 Motivation and Research Question

Three distinct regulatory filing regimes require institutional investors and public officials to disclose their equity holdings: the SEC 13F form (institutional investment managers with  $\geq$  \$100M AUM), the U.S. House Financial Disclosure report (members of Congress), and the U.S. Office of Government Ethics 278e annual disclosure (executive-branch officials). Although these filings differ in granularity and timing, each provides a public record of approximate portfolio weights that can serve as an *investor view* within a quantitative portfolio construction framework.

| Person         | Filing type     | Source                                      |
|----------------|-----------------|---------------------------------------------|
| Warren Buffett | SEC 13F         | Berkshire Hathaway, as of 2025-12-31        |
| Nancy Pelosi   | House FD report | Report 10066169; midpoints of ranges        |
| Donald Trump   | OGE 278e        | Annual disclosure; midpoints / lower bounds |

Range-based values (which appear in both House FD and OGE 278e filings) are converted to midpoints where both bounds are available, and to the reported lower bound otherwise.

Can a Black-Litterman overlay improve portfolio quality relative to:

(1) the **disclosed** portfolio itself (passive replication)?

(2) a **mean-variance** (sample-estimated Markowitz) baseline?

*Improvement* is assessed on annualised Sharpe ratio, Sortino ratio, maximum drawdown, HHI concentration, and average turnover.

## 2 Data Pipeline

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The pipeline ingests two CSV files — holdings disclosures and adjusted daily prices — and produces a `CaseStudyResult` dataclass containing the three `BacktestResult` objects, one per strategy. Figure 1 summarises the flow.

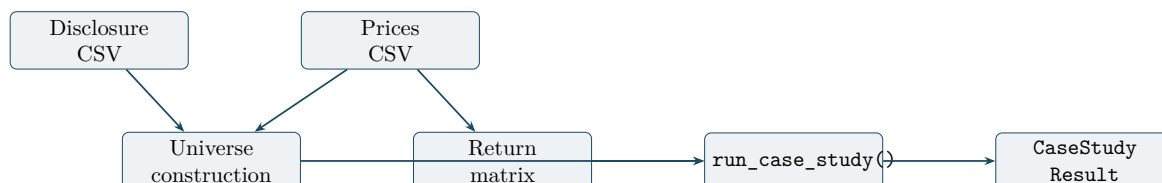


Figure 1: End-to-end data pipeline. Both CSV files are validated on load before any modelling occurs.

### 2.1 Disclosures (`data/disclosures.py`)

`load_disclosures_csv()` schema-validates the input on load, dropping any rows with missing or non-positive weights and emitting a warning for each dropped row.

`latest_portfolio_for_aliases()` selects the most recent filing date for a given set of ticker aliases, ensuring that the disclosed weights reflect the latest available snapshot without introducing any look-ahead bias relative to the price history.

### 2.2 Prices (`data/prices.py`)

The price dataset covers 46 tickers from 2018-01-02 to 2025-12-30 using split- and dividend-adjusted daily closing prices sourced from Yahoo Finance. `load_prices_csv()` detects and warns on duplicate `(date, ticker)` pairs. `to_return_matrix()` pivots the long-format price table into a wide daily simple return matrix aligned to the backtest calendar.

## 3 Three Strategies

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All three strategies share the same asset universe, lookback window, rebalance schedule, and long-only constraint, ensuring that any observed performance differences are attributable to the portfolio construction method alone.

| Strategy               | Description                                                                                                                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Disclosed</b>       | Static buy-and-hold of the filed portfolio weights. No rebalancing occurs; turnover is zero. This represents an upper bound on what a fully passive follower of the public disclosure can achieve.                                           |
| <b>Mean-Variance</b>   | Long-only Markowitz optimisation using the sample mean and covariance estimated over the lookback window. This is the classical quantitative benchmark, known to be sensitive to estimation error in the mean.                               |
| <b>Black-Litterman</b> | Equilibrium prior blended with disclosure-implied views to form a posterior expected return vector and covariance matrix, which are then passed to the same long-only quadratic programme. This is the primary strategy under investigation. |

## 4 Mean-Variance Optimisation (MVO)

### 4.1 Estimation

The mean-variance strategy uses classical Markowitz optimisation driven entirely by the sample moments of the lookback window. Let  $R$  denote the  $T \times n$  matrix of daily simple returns, where  $T$  is the number of trading days in the lookback window and  $n$  is the number of assets in the universe. Each row  $\mathbf{r}_t \in \mathbb{R}^n$  is the vector of simple returns for all  $n$  assets on day  $t$ , so that  $R = [\mathbf{r}_1; \mathbf{r}_2; \dots; \mathbf{r}_T]$ . The sample mean vector and sample covariance matrix are:

$$\hat{\boldsymbol{\mu}} = \frac{1}{T} \sum_{t=1}^T \mathbf{r}_t \in \mathbb{R}^n \quad (1)$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{T-1} \sum_{t=1}^T (\mathbf{r}_t - \hat{\boldsymbol{\mu}})(\mathbf{r}_t - \hat{\boldsymbol{\mu}})^\top \in \mathbb{R}^{n \times n} \quad (2)$$

`estimate_mean_cov()` enforces a minimum of  $T \geq 6$  observations, raising a `ValueError` otherwise. This guard prevents degenerate estimates: with fewer than 6 daily returns the sample mean is dominated by noise and the covariance matrix has at most rank  $T - 1$ , which for very small  $T$  can be far below  $n$  and render the matrix singular. Any remaining NaN entries in  $\hat{\boldsymbol{\Sigma}}$  (arising from tickers that began trading partway through the window) are replaced with zero.

More generally, a well-conditioned covariance estimate requires  $T > n$ ; when  $T \leq n$  the sample covariance is rank-deficient and the linear system in Equation (4) below becomes ill-conditioned. With the current configuration (6-month lookback at daily frequency)  $T \approx 126$  and  $n \leq 22$ , giving a ratio of roughly 6:1 — comfortably above the bare minimum and within the range often recommended in practice ( $T > 3n$  to  $5n$ ).

### 4.2 Long-Only Portfolio Construction

The unconstrained Markowitz tangency portfolio satisfies:

$$\mathbf{w}^* = \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}} \quad (3)$$

In practice, `long_only_markowitz_weights()` solves this system with a small ridge regularisation term added to the diagonal for numerical stability:

$$(\hat{\Sigma} + \delta I) \mathbf{w} = \hat{\boldsymbol{\mu}}, \quad \delta = 10^{-6} \quad (4)$$

The solution is then projected to a long-only, fully-invested portfolio by clipping negative weights to zero and renormalising:

$$w_i = \frac{\max(w_i^*, 0)}{\sum_j \max(w_j^*, 0)}, \quad \sum_i w_i = 1, \quad w_i \geq 0 \quad (5)$$

If all raw weights are non-positive (degenerate solution), the strategy falls back to an equal-weight allocation  $w_i = 1/n$  and logs a warning.

| Symbol                   | Meaning                                                    |
|--------------------------|------------------------------------------------------------|
| $R$                      | $T \times n$ matrix of daily simple returns                |
| $T$                      | Number of trading days in the lookback window              |
| $n$                      | Number of assets in the universe                           |
| $\mathbf{r}_t$           | $n$ -vector of simple returns on day $t$ (row $t$ of $R$ ) |
| $\hat{\boldsymbol{\mu}}$ | $n$ -vector of sample mean returns                         |
| $\hat{\Sigma}$           | $n \times n$ sample covariance matrix                      |
| $\mathbf{w}^*$           | $n$ -vector of unconstrained Markowitz weights             |
| $w_i$                    | Weight of asset $i$ after long-only projection             |
| $\delta$                 | Ridge regularisation scalar (default: $10^{-6}$ )          |
| $I$                      | $n \times n$ identity matrix                               |

The sample mean  $\hat{\boldsymbol{\mu}}$  is notoriously noisy: estimation error in expected returns dominates portfolio selection and often produces extreme, concentrated allocations. This is the central motivation for the Black-Litterman framework, which replaces the sample mean with a Bayesian posterior that shrinks towards an equilibrium prior and is therefore less sensitive to estimation noise.

## 5 The Black-Litterman Model

### 5.1 Inputs

The Black-Litterman model combines two sources of information — an *equilibrium prior* derived from market-clearing weights, and a set of *investor views* derived from the disclosed portfolio — to produce a posterior distribution over expected returns. Both sources share a common asset covariance matrix  $\Sigma = \hat{\Sigma} \in \mathbb{R}^{n \times n}$ , the same sample covariance estimated in the MVO section (Equation 2).

The market-weight vector  $\mathbf{w}_{\text{mkt}} \in \mathbb{R}^n$  is constructed from the disclosed portfolio: the filed dollar weights are re-indexed to the  $n$  valid tickers in the current lookback window (tickers with all-NaN returns are excluded) and renormalised so that  $\mathbf{1}^\top \mathbf{w}_{\text{mkt}} = 1$ . These weights serve as a proxy for the “market portfolio” within the filing’s universe — the allocation that, under equilibrium assumptions, clears the market.

## 5.2 Equilibrium Prior

The framework begins by computing the *implied equilibrium excess return vector* by reverse-engineering the market-clearing condition. Given  $\mathbf{w}_{\text{mkt}}$ , the risk-aversion scalar  $\lambda$ , and the covariance matrix  $\Sigma$ :

$$\boldsymbol{\pi} = \lambda \Sigma \mathbf{w}_{\text{mkt}} \in \mathbb{R}^n \quad (6)$$

with  $\lambda = 2.5$  (default). Intuitively,  $\boldsymbol{\pi}$  answers the question: “What expected returns would make a mean-variance investor willingly hold  $\mathbf{w}_{\text{mkt}}$ ?” These equilibrium returns serve as the prior mean for the distribution of expected returns.

## 5.3 Views from Disclosures

The disclosed portfolio weights are used to construct a set of investor *views* about expected asset returns. In this implementation the view-picking matrix  $P$  is set to the  $n \times n$  identity matrix:

$$P = I_n \in \mathbb{R}^{n \times n} \quad (7)$$

This means there is one *absolute view* per asset — each view states a belief about the expected return of a single asset rather than a relative comparison between assets. The corresponding view return vector  $\mathbf{q} \in \mathbb{R}^n$  is set to the sample mean return vector  $\hat{\boldsymbol{\mu}}$  from the lookback window (Equation 1):

$$\mathbf{q} = \hat{\boldsymbol{\mu}} \quad (8)$$

Since the return matrix  $R$  contains daily simple returns and the default lookback window spans 6 months ( $T \approx 126$  trading days), each element  $q_i = \hat{\mu}_i$  is a *mean daily simple return* — typically on the order of  $10^{-4}$  — not an annualised or monthly figure. The uncertainty in these views is modelled as proportional to the prior covariance, scaled by the confidence parameter  $c \in (0, 1)$ :

$$\boldsymbol{\Omega} = \frac{1-c}{c} P \Sigma P^\top \in \mathbb{R}^{n \times n} \quad (9)$$

The parameter  $c$  (default: 0.65) controls the relative weight given to the views versus the equilibrium prior. As  $c \rightarrow 1$  the view uncertainty  $\boldsymbol{\Omega} \rightarrow 0$  and the posterior collapses to the sample-mean view; as  $c \rightarrow 0$  the uncertainty grows without bound and the posterior collapses to the equilibrium prior  $\boldsymbol{\pi}$ .

## 5.4 Posterior Update

Following He and Litterman (1999), the posterior is derived analytically. Define the posterior precision matrix:

$$M = (\tau \Sigma)^{-1} + P^\top \boldsymbol{\Omega}^{-1} P \in \mathbb{R}^{n \times n} \quad (10)$$

where  $\tau = 0.05$  (default) is a scalar that controls the uncertainty in the equilibrium prior — it scales the prior covariance so that  $\tau \Sigma$  represents the investor’s uncertainty about the true mean returns (as distinct from  $\Sigma$  itself, which represents the uncertainty in realised returns). The posterior mean and covariance are:

$$\boldsymbol{\mu}_{\text{BL}} = M^{-1} [(\tau \Sigma)^{-1} \boldsymbol{\pi} + P^\top \boldsymbol{\Omega}^{-1} \mathbf{q}] \in \mathbb{R}^n \quad (11)$$

$$\Sigma_{\text{BL}} = \Sigma + M^{-1} \in \mathbb{R}^{n \times n} \quad (12)$$

The posterior mean  $\boldsymbol{\mu}_{\text{BL}}$  is a precision-weighted blend of the equilibrium prior  $\boldsymbol{\pi}$  and the sample-mean views  $\mathbf{q}$ : the more confident the views (higher  $c$ , smaller  $\boldsymbol{\Omega}$ ), the closer  $\boldsymbol{\mu}_{\text{BL}}$  sits

to  $\mathbf{q}$ ; the less confident, the closer it sits to  $\boldsymbol{\pi}$ . The posterior covariance  $\boldsymbol{\Sigma}_{\text{BL}}$  is always larger than  $\boldsymbol{\Sigma}$  alone, reflecting the additional estimation uncertainty introduced by the Bayesian blending step.

The pair  $(\boldsymbol{\mu}_{\text{BL}}, \boldsymbol{\Sigma}_{\text{BL}})$  is then passed to the same long-only solver used in MVO (Equations 4–5) with the full-investment constraint  $\mathbf{1}^\top \mathbf{w} = 1$ ,  $\mathbf{w} \geq 0$ . If the solver returns all non-positive weights the strategy falls back to equal-weight allocation and logs a warning.

| Symbol                            | Meaning                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------|
| $\boldsymbol{\Sigma}$             | $n \times n$ sample covariance matrix (same as $\hat{\boldsymbol{\Sigma}}$ from MVO) |
| $\mathbf{w}_{\text{mkt}}$         | $n$ -vector of disclosed portfolio weights (renormalised to valid tickers)           |
| $\lambda$                         | Risk-aversion scalar (default: 2.5)                                                  |
| $\boldsymbol{\pi}$                | $n$ -vector of implied equilibrium excess returns                                    |
| $P$                               | $n \times n$ view-picking matrix (identity: one absolute view per asset)             |
| $\mathbf{q}$                      | $n$ -vector of view returns (set to sample mean $\hat{\boldsymbol{\mu}}$ )           |
| $c$                               | View confidence scalar (default: 0.65; swept in Notebook 03)                         |
| $\boldsymbol{\Omega}$             | $n \times n$ view uncertainty matrix                                                 |
| $\tau$                            | Prior uncertainty scalar (default: 0.05)                                             |
| $M$                               | $n \times n$ posterior precision matrix                                              |
| $\boldsymbol{\mu}_{\text{BL}}$    | $n$ -vector of BL posterior expected returns                                         |
| $\boldsymbol{\Sigma}_{\text{BL}}$ | $n \times n$ BL posterior covariance matrix                                          |

## 6 Walk-Forward Backtest Engine

### 6.1 Design and Lookahead-Bias Prevention

`rolling_backtest()` in `backtest/engine.py` implements a strict no-lookahead-bias design: at each month-end rebalance date  $t$ , portfolio weights are computed using only the 6-month trailing window of returns ending at  $t$ , and those weights are applied to returns *starting from the period after  $t$* . The `weight_history` output is indexed by decision date, making it straightforward to audit the lag.

### 6.2 Configuration

All hyperparameters are declared in `configs/case_studies.yaml` and loaded into a frozen `BacktestConfig` dataclass at runtime. No magic numbers appear in the source code.

| Parameter                        | Value           |
|----------------------------------|-----------------|
| <code>lookback_periods</code>    | 6 months        |
| <code>rebalance_frequency</code> | Month-end (ME)  |
| <code>risk_aversion</code>       | $\lambda = 2.5$ |
| <code>tau</code>                 | $\tau = 0.05$   |
| <code>view_confidence</code>     | $c = 0.65$      |

### 6.3 Output

`rolling_backtest()` returns a `BacktestResult` dataclass with three fields: `.returns` (daily return series), `.nav` (cumulative NAV indexed from 1.0), and `.weight_history` (DataFrame of rebalance-date weights).

## 7 Performance Metrics

All metrics are computed by `backtest/metrics.py`. The helper `infer_periods_per_year()` automatically detects daily, monthly, or quarterly frequency from the return index and emits a `UserWarning` when fewer than three observations are available.

| Metric            | Notes                                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annual return     | Geometric compounding from NAV series.                                                                                                                                                            |
| Annual volatility | Annualised standard deviation of daily returns.                                                                                                                                                   |
| Sharpe ratio      | Excess return over annualised volatility; risk-free rate defaults to zero.                                                                                                                        |
| Sortino ratio     | Excess return over downside deviation; returns <code>nan</code> (not $+\infty$ ) when no negative daily return exists in the window, avoiding misleading statistics on strongly trending periods. |
| Max drawdown      | Maximum peak-to-trough decline in cumulative NAV.                                                                                                                                                 |
| HHI concentration | $\sum_i w_i^2$ ; ranges from $1/n$ (equal-weight) to 1.0 (fully concentrated).                                                                                                                    |
| Average turnover  | Mean $L_1$ weight change per rebalance:<br>$\frac{1}{T} \sum_t \ \mathbf{w}_t - \mathbf{w}_{t-1}\ _1$ .                                                                                           |

`summarize_strategy()` aggregates all seven metrics into a dict, which the notebooks collect into a combined summary DataFrame for side-by-side comparison across strategies and persons.

## 8 Analysis Notebooks

The repository ships four Jupyter notebooks. All notebooks iterate over the **Buffett**, **Pelosi**, and **Trump** case studies from a shared `results` dict — no person keys are hardcoded.

### 8.1 Notebook 01 — Data Quality & Universe Overview

Performs schema checks on the raw holdings and price CSVs, reports holdings coverage (fraction of disclosed tickers with matching price history), and renders portfolio composition pie charts for each person. The output provides a sanity check that the data pipeline has ingested the filings correctly before any modelling is run.

### 8.2 Notebook 02 — Strategy Comparison Case Study

The main results notebook. For each person it plots:



- Equity curves (cumulative NAV) for all three strategies on a shared axis.
- Drawdown time series highlighting the 2020 COVID drawdown and 2022 rate-shock episodes.
- Black-Litterman weight evolution over rebalance dates, showing how the posterior allocation shifts as the disclosure-implied view interacts with the changing equilibrium prior.
- A combined summary table with all seven metrics for all three strategies.

### 8.3 Notebook 03 — Black-Litterman Sensitivity Analysis

Sweeps the view confidence parameter  $c \in \{0.20, 0.25, \dots, 0.95\}$  (16 values) across all three persons, running 48 independent backtests in total. The results are displayed in a  $2 \times 3$  subplot grid showing how annualised return, volatility, Sharpe ratio, HHI, and average turnover respond to  $c$ .

Key findings from the sweep:

- The Sharpe ratio is *non-monotonic* in  $c$  on real data — the value of  $c$  that maximises Sharpe differs across persons and cannot be determined a priori.
- Higher confidence causes turnover to increase when the disclosed view diverges substantially from the equilibrium prior.
- HHI spikes near extreme values of  $c$ , indicating that very high or very low confidence produces concentrated allocations.

The best-by-Sharpe row is printed for each person, providing a principled starting range for `view_confidence` before any out-of-sample reporting.

### 8.4 Notebook 04 — Benchmark Attribution & Alpha Decomposition

Fits an OLS factor model to each strategy's return series:

$$r_{\text{strat},t} = \alpha + \beta_1 r_{\text{SPY},t} + \beta_2 r_{\text{Tech},t} + \beta_3 r_{\text{Energy},t} + \beta_4 r_{\text{Fin},t} + \beta_5 r_{\text{CD},t} + \varepsilon_t \quad (13)$$

where benchmarks are sourced preferentially from ETF tickers (SPY, XLK, XLE, XLF, XLY) with an automatic fallback to ticker-proxy averages (e.g. AAPL + MSFT + NVDA for Tech) if the ETF is absent from the price file. A provenance table is printed per person.

For each person the notebook produces:

- A  $\hat{\beta}$  bar chart (positive loadings in teal, negative in red).
- An arithmetic return contribution decomposition:  $\bar{r}_{\text{ann}} \approx \hat{\alpha} \cdot f + \sum_k \hat{\beta}_k \bar{r}_{k,\text{ann}} + \text{residual}$ , where  $f$  is periods per year.
- A return correlation heatmap between the strategy and each factor.
- Rolling 12-month  $\hat{\alpha}$  and  $\hat{\beta}$  versus SPY, revealing regime shifts hidden in the full-sample statistics — most visibly during the 2020 COVID drawdown and the 2022 rate shock.

## 9 Codebase Architecture

### 9.1 Package Layout

The `portfolio_bl` package lives entirely under `src/` and is organised into three sub-packages.

| Module                                 | Responsibility                                                                                 |
|----------------------------------------|------------------------------------------------------------------------------------------------|
| <code>config.py</code>                 | YAML loading; <code>BacktestConfig</code> and <code>CaseStudyConfig</code> frozen dataclasses. |
| <code>pipeline.py</code>               | <code>run_case_study()</code> orchestrator; returns <code>CaseStudyResult</code> .             |
| <code>data/disclosures.py</code>       | Load, validate, and filter holdings CSV.                                                       |
| <code>data/prices.py</code>            | Load prices CSV; pivot to return matrix.                                                       |
| <code>models/black_litterman.py</code> | <code>implied_equilibrium_returns()</code> ; <code>black_litterman_posterior()</code> .        |
| <code>models/mean_variance.py</code>   | <code>long_only_markowitz()</code> with equal-weight fallback.                                 |
| <code>backtest/engine.py</code>        | <code>rolling_backtest()</code> walk-forward loop.                                             |
| <code>backtest/metrics.py</code>       | All seven performance metrics; <code>summarize_strategy()</code> .                             |

### 9.2 Quality and Observability

- **63 tests** (unit and integration) runnable with `pytest -tb=short`. Tests cover the BL posterior algebra, backtest engine edge cases (empty windows, all-positive returns), and metric correctness.
- `logging.getLogger(__name__)` is configured in every module. The `-verbose` flag elevates the log level to `DEBUG`, printing per-rebalance diagnostics without modifying any source file.
- `FileNotFoundError` and `ValueError` are raised with descriptive messages on bad or missing inputs; no silent failures.
- `__all__` is defined in every `__init__.py`, making the public API explicit.
- All BL hyperparameters are exposed via YAML; no magic numbers appear in the source code.

### 9.3 CLI Quick-Start

```
PYTHONPATH=src python scripts/run_case_study.py \
  --config configs/case_studies.yaml \
  --person buffett \
  --verbose
```

Swapping `-person buffett` for `pelosi` or `trump` runs the identical pipeline on a different case study. All three persons can also be run programmatically by calling `run_case_study()` in a loop with different `CaseStudyConfig` objects.

## 10 Known Limitation and Extensions

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The BL posterior assumes multivariate Gaussianity in both returns and views. Equity return distributions exhibit significant excess kurtosis and left-tail asymmetry, particularly during stress periods directly visible in this backtest — the **2020 COVID drawdown** and the **2022 rate shock**. A production-grade system would replace the Gaussian prior with Meucci's Entropy Pooling / Copula Opinion Pooling framework, which separates the marginal distributions from the dependence structure and allows views to be expressed on arbitrary statistics including implied volatility.

The practical consequences within this backtest are:

- The posterior covariance  $\Sigma_{BL}$  underestimates tail risk during crisis periods, causing the optimiser to over-allocate to assets that appear low-volatility under normality but exhibit fat tails under stress.
- The Sortino ratio and maximum drawdown are therefore the most informative diagnostics in this framework, as they directly capture downside behaviour without assuming symmetry.

The Entropy Pooling approach is the natural upgrade path: it preserves the Bayesian blending intuition of Black-Litterman while removing the Gaussian assumption entirely.

## 11 Summary

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### What the framework provides

- End-to-end walk-forward backtest on real, publicly-filed portfolio data — not synthetic or simulated holdings.
- Fair three-way strategy comparison: Disclosed, Mean-Variance, and Black-Litterman on an identical universe and schedule.
- Full sensitivity sweep over the key BL hyperparameter ( $c$ ), producing 48 backtests across three persons.
- OLS factor attribution with rolling 12-month diagnostics.
- YAML-configurable — swap in any universe or parameter set without modifying source code.
- 63 tests enforcing correctness at every layer of the stack.

### Key design decisions

- No-lookahead-bias by construction in the engine.
- Graceful degradation to equal-weight with a logged warning rather than a hard crash.
- `nan` (not  $+\infty$ ) Sortino on all-positive return periods.
- Real adjusted prices (2018–2025, 46 tickers) via Yahoo Finance.